

Vroomie

Audio Detection Pipeline

Technical Specification

Complete engineering reference for the two-path detection engine

Derived from the shipped source · every constant verified in code
github.com/DGeorgeA/vroomie-app · branch main
Live at vroomie.in

All performance figures herein are measured results on held-out data, reproducible with the committed test harnesses.

it is ready, so a cold-cache fetch no longer costs the opening seconds of a session.

3. Path A — constellation fingerprinting

Identifies *known reference recordings*. Self-contained module; no shared thresholds with Path B.

3.1 Parameters

Constant	Value	Meaning
NFFT	1024	64 ms window @ 16 kHz, 31.25 Hz/bin
HOP	256	16 ms frame advance
BANDS	[0, 20, 40, 80, 160, 320, 512]	six log-spaced bin ranges
PEAK_FACTOR	1.6	peak must exceed its band mean by this
FANOUT	6	targets paired per anchor peak
DT_MIN/DT_MAX	1 / 48	pairing window in frames (16–768 ms)
DF_MAX	160	max frequency delta in bins

3.2 Algorithm

- Spectrogram** — Hann-windowed 1024-point radix-2 FFT per frame, magnitude spectrum over 512 bins.
- Peak picking** — per frame, per band, the strongest bin is retained if it exceeds that band's mean by **PEAK_FACTOR**. Peaks are *relative* maxima, which is what makes the representation invariant to level and to broad EQ changes.
- Hashing** — each anchor peak pairs with up to **FANOUT** later peaks inside the (**DT**, **DF**) window. Each pair packs into a single 32-bit integer:

$$\text{hash} = (\text{f1} \ \& \ 0\text{x1FF}) \ll 15 \mid ((\Delta\text{f} + 256) \ \& \ 0\text{x1FF}) \ll 6 \mid (\Delta\text{t} \ \& \ 0\text{x3F})$$

The anchor's absolute frame index is stored alongside.

- Matching by time-offset coherence** — for each query hash present in the index, every occurrence votes for the pair (**referenceId**, **referenceTime** – **queryTime**). A genuine match aligns the *entire* recording, so votes pile into one offset bin; unrelated audio scatters them. The score is the height of the tallest single-offset bin.

This coherence requirement is the reason the path can be sensitive without being trigger-happy: unrelated audio cannot manufacture a consistent time alignment across hundreds of independent hashes.

3.3 Decision

`normalised = score / min(queryHashCount, ref.hash_count)`

`instant : score ≥ 600 AND normalised ≥ 0.050`

`sustained : score ≥ 480 AND normalised ≥ 0.042`

`AND the same reference wins 2 consecutive attempts (~0.9 s apart)`

`matched = instant OR sustained`

Why `min()` in the denominator. Dividing by query hashes alone makes any reference shorter than the listen window structurally incapable of reaching the threshold. Measured: a genuine replay of an indexed 1.5 s reference peaked at 0.035 against a 0.05 requirement — it failed against *itself*. The artifact therefore ships

`hash_count` per reference.

Why a sustained tier exists. The 600 threshold was derived from a simulated speaker channel where positives scored 621+. Real phone-through-speaker telemetry measured correct identifications at 494–553 — right labels, rejected by an over-tight bar. Lowering the bar alone was unsafe (the worst healthy negative measured 443), so the lower tier is paired with persistence: a real replay retains its identity across independent windows, a coincidental peak alignment does not.

`LISTEN_SECONDS = 5` (rolling ring buffer), `MIN_LISTEN_SECONDS = 3` (no attempt before this much audio).

3.4 Index

`public/constellation_v1.json` — 8 references, 155,288 entries, 1.6 MB, service-worker cached. Stored as two parallel base64 `Int32Arrays` (`keys` sorted, `vals` packed as `refId << 20 | frameIndex`), hydrated at runtime into a `Map<hash, Int32Array>` using `subarray` views — no per-entry object allocation.

Reference	Hashes
Alternator bearing noise	22,318
MotorStarter	22,320
Piston	22,236
PowerSteeringPump	22,291
RockerArmAndValve	22,186
SerpentineBelt	22,250
Intake leak	10,899
Timing chain rattle	10,788

Two deliberate exclusions, both measured: `water_pump_failure_critical.wav` is a synthetic sine tone rather than a recording; `misfire_detected_medium.wav` is acoustically an *irregular idle* and produced the only two healthy false fires observed (496 and 523, sustained across four consecutive attempts, so persistence could not separate them). Misfire detection continues on Path B, which separates it using healthy anchors. A `MIN_REF_SECONDS = 4.0` floor also excludes the 1.5 s power-steering variants.

Cost. `tryMatch()` measures 22 ms median on desktop; roughly 8× headroom against the 900 ms cadence even assuming a phone 5× slower.

4. Path B — YAMNet embedding matching

Generalises to vehicles never seen before. Model: YAMNet via TensorFlow.js, executed entirely on-device, producing per-frame 1024-d embeddings and 521-class scores, mean-pooled across the 1 s window.

4.1 Stage 1 — acoustic domain gate

Decides whether the window is vehicle audio *before* any fingerprint comparison. Uses the 521 class scores, which the earlier architecture computed and discarded.

- `VEHICLE_MECH_INDICES` — 53 mechanical/vehicle classes (Engine, Idling, Rattle, Gears, Whir, Clatter, ...). `Mechanical fan` and `Air conditioning` are deliberately **excluded**: household fans passed the gate and matched hiss-like references.

- **INTERFERER_INDICES** — the contiguous human-sound and music ranges plus Television, Radio, Silence, Whistling, Whistle.

```
accepted = VEHICLE_MECH_INDICES.has(top1)
|| (!isInterferer && interfererScore < 0.15) // GENERIC_INTERFERER_CEILING
|| (vehicleScore ≥ 0.03 && vehicleScore > interfererScore) // VEHICLE_SCORE_FLOOR
|| (isInterferer && vehicleScore ≥ 0.02 && interfererScore ≤ 0.30) // speaker rescue
```

The **generic-acoustics clause** exists because real faults frequently do not classify as vehicle sounds: measured, alternator bearing → "White noise"/"Waterfall", misfire → "Explosion"/"Rain", pump whine → "Sine wave". A strict vehicle-only gate discarded those windows entirely.

The **speaker-rescue clause** (v9.9) was added from field diagnosis: playback through a real speaker shifts YAMNet's top-1 toward Speech/Applause/Bell, and the previous hard interferer veto discarded windows that simultaneously carried vehicle evidence. Boundary measured on 70 vetoed fault windows vs 108 real speech/music windows — recovers 29, admits 0. Real speech sits at interferer 0.47–0.98 with vehicle evidence ≈ 0.001 , far outside the clause.

4.2 Stage 2 — discriminative match

```
bestFault = max cosine(liveEmbedding, faultRef) over 352 fault embeddings
bestAnchor = max cosine(liveEmbedding, anchor) over 94 anchors
margin = bestFault - bestAnchor

candidate bestFault ≥ 0.45 (ANOMALY_THRESHOLD) AND margin ≥ 0.04 (ANCHOR_MARGIN)
near bestFault ≥ 0.45 AND 0.02 ≤ margin < 0.04 (NEAR_ANCHOR_MARGIN)
```

The **anchors are the load-bearing invention**. YAMNet embeddings of any two sustained sounds sit at 0.7–0.9 cosine, so "nearest fault" is meaningless on its own. Measured: a healthy idling car scores **0.892–0.918** against power-steering references — higher than several genuine faults. The anchor set supplies the "compared to what?": 78 healthy engine anchors (idle, startup, braking; even-numbered Kaggle clips only, odd held out) plus 16 interferer anchors (four real speech recordings, white/pink noise at three levels, fan simulation, 440 Hz tone, synthetic music, traffic).

Near-band windows still count as **clean** — their **status** and **reason** are unchanged — so the primary rule is provably unaffected by the near tier's existence.

4.3 Stage 3 — session decision

In `AudioRecorder.jsx`, evaluated at stop:

```
accepted = cleanWindows + candidateWindows

PRIMARY : one fault FAMILY holds ≥ 45% of accepted windows, min 3 accepted
RECOVERY : nothing confirmed, but total candidates ≥ 60% of accepted
           → plurality family; on a hit-tie the leader's summed raw similarity
           must exceed the runner-up by ≥ 1.10x, else decline
NEAR : nothing confirmed, and one family holds ≥ 85% of accepted windows
       in the near band → "POSSIBLE – VERIFICATION REQUIRED", 65–69%,
       severity capped at 'medium'
```

Votes aggregate by **fault family** (**fault_type**), not by individual label: three separate alternator-bearing references once split a 57%-candidate session three ways and confirmed nothing.

The recovery vote handles acoustic distance splitting one fault across two families — measured, a rocker arm at 2 m produced 8 of 12 confident candidate windows split 4/4 and reported nothing. The tie-break ranks by **raw similarity sum, not margin**: margin ranking picked the wrong family in 5 of 9 tied probe runs.

4.4 Confidence

```
marginToConfidence(margin) = clamp(0.70 + 1.08 × (margin - 0.04), 0.70, 0.97)
```

Derived from the **margin** — the match strength after subtracting the best healthy/noise anchor — not from raw similarity, which sits at 0.7–0.9 for any two sustained sounds and previously produced meaningless "80% confidence" claims. A confirmed anomaly is therefore $\geq 70\%$ by construction. Fingerprint hits report a fixed 97%; the near tier maps to 65–69%, strictly below the confirmed floor so the tiers cannot blur.

Published as: *"There is a 77% possibility that there could be a possible Piston (Piston.wav)".*

5. Reference artifact factory

`public/fingerprints_v9.json` — 1024-d, int8-quantised ($\text{value} = \text{min} + \text{byte} \times \text{scale}$), 352 fault embeddings, 94 anchors. Built offline; **never generated in the browser** (that cost every client an 11 MB download and minutes of startup).

Pipeline per reference: QC (reject synthetic tones by YAMNet dominant class, clips < 1 s, near-silent) → mono → 16 kHz → RMS-normalise to 0.05 → chunk selection → 8 augmentation variants → YAMNet → quantise.

Variant	Purpose	Count
<code>orig</code>	baseline	46
<code>band</code>	phone-band filter	43
<code>noise</code>	+15 dB SNR noise	44
<code>rate+ / rate-</code>	$\pm 2\%$ rate shift	43 / 46
<code>echo</code>	room reflection	43
<code>speaker</code>	gentle speaker replay	44
<code>phonespk</code>	harsh phone-speaker channel	43

Per-family cap of 100 prevents dataset dominance. Before it, `power_steering` held 376 of 560 embeddings (58.8%) and absorbed 8 of 9 healthy false alarms. Current distribution: alternator 100, power_steering 100, and 24 each for intake_leak, misfire, piston_knock, rocker_valve, serpentine_belt, timing_chain; motor_starter 8.

Critical invariant: live windows and reference chunks must undergo mathematically identical preprocessing. Both paths mono → 16 kHz → RMS-normalise to 0.05 → 1 s window. Measured parity: identical RMS after normalisation (0.05000 both), identical dimensions (1024 / 521); the residual embedding cosine of 0.9528 between direct and channel-simulated audio is the acoustic channel itself, not a preprocessing mismatch. There is **no FFT/Mel/MFCC stage** in Path B — YAMNet consumes normalised PCM directly.

6. Failure handling and diagnosability

Every abort path persists a hidden `status: 'rejected'` row (filtered from all history UIs) carrying: window counts, silence-vs-domain rejection split, the top YAMNet classes the audio was heard as, the device's *applied* capture settings, the best fingerprint score even when it never cleared threshold, and the engine build.

Three abort conditions, each with a specific user-facing message — never a fabricated "healthy" result:

- zero windows analysed (model not ready / session too short)
- reference artifact failed to load
- majority of windows rejected → *"heard as `class` ... N quiet / M non-vehicle of K windows · closest fingerprint `label` `score`/600"*

A fingerprint hit bypasses all three: the audio *was* identified, so an "unable to detect" message would be false.

This telemetry is what diagnosed the 13 August demo failure two days later without device access — fingerprint scores of 79/167/88 against 549–736 for identical audio, with 37 of 42 windows still accepted, isolating Zoom's acoustic echo cancellation as the cause.

7. Measured performance

Measure	Result	Harness
Stress matrix — 8 refs × 4 channels × 4 volumes × 3 offsets	384/384 (100%)	<code>stress_detection_matrix.mjs</code>
Fingerprint false fires — 140 held-out healthy clips	0	<code>verify_sustained_tier.mjs</code>
Fingerprint false fires — 9 interferers	0 (worst 56)	<code>verify_sustained_tier.mjs</code>
Embedding healthy false alarms	6/140 (4.3%)	<code>rca_healthy_fp_sweep.mjs</code>
Held-out unseen fault recall (embedding)	~2 in 3	<code>benchmark_discrimination.mjs</code>
Fingerprint latency	~3.7 s	stress matrix
<code>tryMatch</code> cost	22 ms median	timing probe

8. Operating constraints

- Path A recognises **exact recordings**, not fault categories — the same limitation that lets Shazam identify a studio track but not a cover. Real-vehicle coverage rests entirely on Path B.
- Nine fault families are supported; injector tick and exhaust leak have no reference audio and cannot be detected at all.
- `intake_leak` replay is partially suppressed by the broadband-noise anchors that provide fan immunity — an intake leak *is* a hiss. A genuine tension, not a defect.
- Devices that force noise suppression may attenuate signatures; now visible per session in `capture_settings`.
- Both accuracy figures in §7 are bounded by reference-library size, not by algorithm design.